

Machine Learning Model Comparison Using scikit-learn

Instructor: Dr. Ruby

Date: April 2026

Datasets used

1. Iris dataset (built into scikit-learn)
2. Heart Disease dataset (303-row classification dataset)

Summary:

This project applied three supervised learning algorithms - Logistic Regression, k-Nearest Neighbors (KNN), and Decision Tree - to two classification datasets: Iris and Heart Disease. The workflow included exploration, preprocessing, train/test splitting, hyperparameter tuning with GridSearchCV, and model evaluation using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC. On the Iris dataset, all models performed strongly, with KNN producing the best test accuracy at 96.67%. On the Heart dataset, KNN produced the best test accuracy at 90.16%, while Logistic Regression remained highly competitive and delivered a ROC-AUC of 0.9589. Overall, the results show that model performance depends on both the structure of the dataset and the way the features are prepared before training.

Dataset Selection:

The Iris dataset was selected because it is a standard benchmark for multiclass classification. According to the scikit-learn documentation, it contains 150 samples, 4 numeric features, and 3 classes, making it useful for comparing basic classifiers on a clean dataset. The Heart Disease dataset was selected because it is more realistic and clinically relevant. The UCI Machine Learning Repository notes that the widely used heart disease database includes 303 instances and focuses on predicting the presence or absence of disease from a subset of attributes that researchers commonly use. In the heart dataset used here, the target variable was AHD, and categorical fields such as ChestPain and Thal required one-hot encoding before model training.

Model Development:

The project used an 80/20 train/test split with stratification to preserve class balance. For Iris, the features were numeric, so median imputation and standardization were enough for preprocessing. For the Heart dataset, numeric columns were imputed and scaled for Logistic Regression and KNN, while categorical columns were one-hot encoded. Decision Tree was trained with a separate preprocessing path that preserved unscaled numeric values. The three chosen algorithms represent different model families: Logistic Regression provides a simple linear baseline, KNN is a distance-based nonparametric method, and Decision Tree is a nonlinear rule-based model. This combination made it easier to compare how different modeling assumptions behave across an easy dataset and a more difficult real-world dataset.

Hyperparameter Tuning:

GridSearchCV was used for all six dataset-model combinations. On Iris, the best Logistic Regression model used $C=1$, the best KNN model used 3 neighbors with distance weighting, and the best Decision Tree used the default unrestricted depth with a minimum split of 2. On the Heart dataset, the best Logistic Regression model used $C=0.1$, the best KNN model used 5 neighbors with uniform weighting, and the best Decision Tree used entropy with $\text{max_depth}=3$. These tuning results show that even simple changes in model settings can improve performance and reduce weak parameter choices.

Results and Evaluation:

Table 1 summarizes the tuned model performance on the Iris dataset.

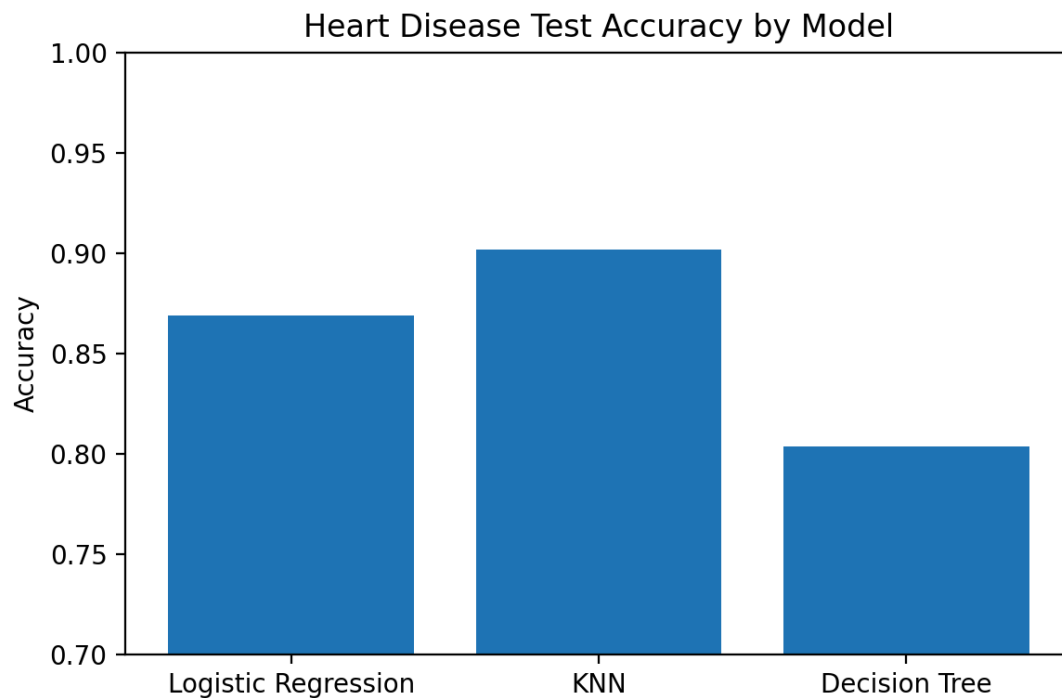
Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV Acc.
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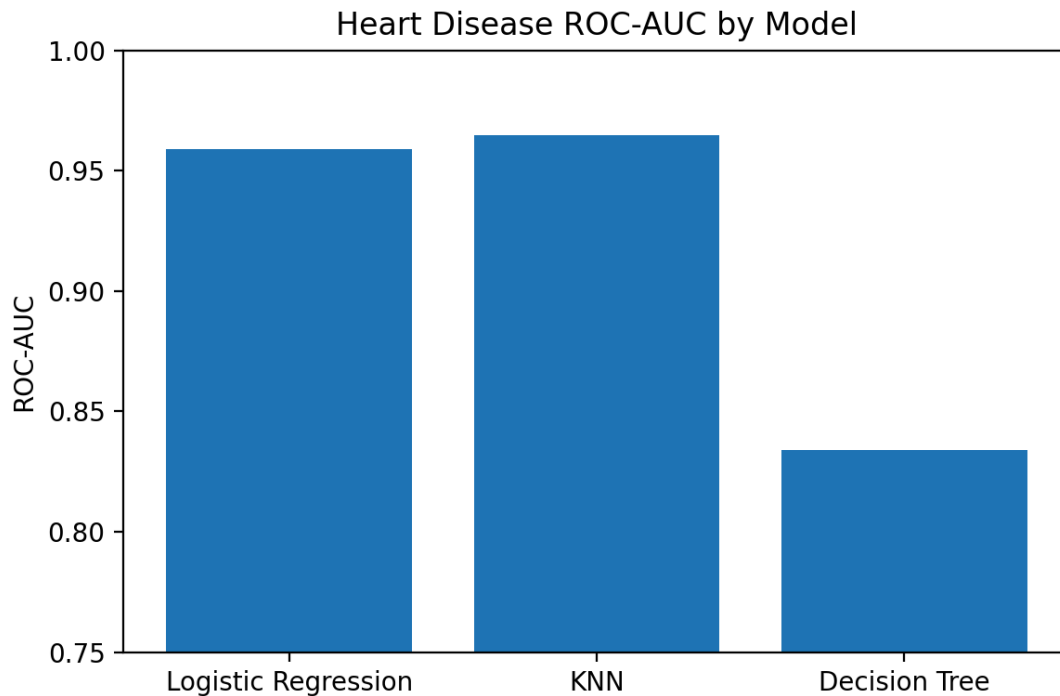
Logistic Regression	0.9333	0.9333	0.9333	0.9333	0.9967	0.9583
KNN	0.9667	0.9697	0.9667	0.9666	1.0	0.9667
Decision Tree	0.9333	0.9333	0.9333	0.9333	0.95	0.9417

Table 2 summarizes the tuned model performance on the Heart dataset.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV Acc.
Logistic Regression	0.8689	0.8766	0.8689	0.869	0.9589	0.8305
KNN	0.9016	0.9096	0.9016	0.9017	0.9648	0.8139
Decision Tree	0.8033	0.8056	0.8033	0.8036	0.8339	0.7686

The Iris results were uniformly strong, which is not surprising because the dataset is small, balanced, and well separated. KNN performed best on Iris with 96.67% test accuracy and a perfect one-vs-rest ROC-AUC of 1.0000. On the Heart dataset, KNN again had the highest test accuracy at 90.16%, while Logistic Regression remained very competitive and produced a ROC-AUC of 0.9589. Decision Tree was the weakest model on the Heart data, suggesting that the dataset was less forgiving and more likely to expose variance and overfitting problems in tree-based splits.





Literature Comparison

A recent Scientific Reports study compared Logistic Regression, SVM, KNN, and Random Forest on an open heart disease dataset using cross-validation and reported that Logistic Regression and KNN each reached 81.9% baseline accuracy, while Random Forest achieved the strongest macro precision, recall, and F1 performance overall. The same study reported average 5-fold cross-validation accuracy around 83.81% for Logistic Regression and 84.15% for KNN. Compared with that study, the current project produced a slightly higher held-out test accuracy for KNN (90.16%) and a higher ROC-AUC for both KNN (0.9648) and Logistic Regression (0.9589), but the comparison is not perfectly direct because the preprocessing details, train/test partition, and exact dataset version are not identical. Even so, both projects point to the same general conclusion: KNN and Logistic Regression are strong classical baselines for heart disease classification, while more complex or ensemble approaches may improve robustness on some metrics.

Conclusion

This project successfully demonstrated the full supervised learning workflow using scikit-learn. The experiments showed that simple models can still perform very well when the data is prepared correctly. The Iris dataset was easy for all three models, while the Heart dataset was more challenging and created clearer separation between model quality. Based on this project, KNN was the strongest model by test accuracy on both datasets, while Logistic Regression stood out as the most stable and interpretable performer on the Heart dataset. A practical limitation is the small size of both datasets, especially the heart dataset, which can make results sensitive to splits and parameter choices. In future work, the project could be expanded with additional datasets, repeated cross-validation, or ensemble methods such as Random Forest or Gradient Boosting.

References

Rimal, Y., Gyawali, P., Sharma, A., & Kharel, M. (2025). Comparative analysis of heart disease prediction using logistic regression, SVM, KNN, and random forest with cross-validation for improved accuracy. *Scientific Reports*, 15, Article 93675.

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